JPX CNN-LSTM
ReLU / Pooling Removal Report

42-observation window | kernel 3 | 8 folds x 20 epochs | seed 20261016
Report date: 2026-10-20 (Asia/Taipei)

8 / 8	160	958
folds completed	epochs completed	evaluation dates

Research question: Can removing both the post-convolution ReLU and MaxPool1d improve Rank IC and Sharpe simultaneously?
Result: Rank IC increased, but Sharpe and the gross additive spread sum fell. This run did not support the hypothesis that both metrics would improve.

Overall result: better rank correlation, weaker portfolio performance

Metric	With ReLU / Pool	Both removed	Change (run - original)
Mean daily Rank IC	0.003754361	0.004774635	+0.001020274
Unannualized pooled Sharpe	0.099446346	0.054660544	-0.044785802
Sample-weighted MAE	0.016032527	0.016026605	-0.000005922
Gross additive spread sum (pp)	23.595058	13.962599	-9.632459

Rank IC rose by 27.18%; Sharpe fell by 45.04%; the gross additive spread sum declined by 9.63 percentage points. MAE improved by only 0.03694% and remained 0.48012% above the zero-prediction MAE. The small MAE improvement does not offset the deterioration in ranking-portfolio performance.

This run's MAE was reconstructed by weighting the eight fold MAEs by sample count, and Earnings by summing the folds. Sharpe and Rank IC come directly from the completion summary. The baseline was checked against the project's pinned configuration, evaluation protocol, and result files. [L1, B1-B3, S1]

Research judgment

Keep the version with ReLU / Pooling as the current primary control, and retain the removal version as a candidate for further validation. This is a historical-path comparison with one seed. CPU versus CUDA, package versions, and two components changed together remain confounded; neither causality nor statistical significance can be claimed.

pp means percentage points. Here it is the sum of daily long-short spreads before costs, not a compounded asset return. This report added no training and changed no project code.

01 Experiment setup and architecture

Checked against the uploaded model, pinned model_2, and run_no_relu.py to avoid mixing in later project settings. [B1, S1-S3]

Baseline - left to right

42 x 6 inputs Conv 6 to 64 kernel 3: 42 to 40	ReLU + Pool(2) 40 to 20 steps	Dropout1d 0.2 3-layer LSTM hidden size 100	Last time step Dropout 0.2 Linear 100 to 1

This run - left to right

42 x 6 inputs Conv 6 to 64 kernel 3: 42 to 40	Retain Conv output 40 steps	Dropout1d 0.2 3-layer LSTM hidden size 100	Last time step Dropout 0.2 Linear 100 to 1

Removing MaxPool1d increased the LSTM input from 20 to 40 time steps; ReLU's clipping of negative values was also removed. The LSTM still contains nonlinear operations, so the overall model did not become linear. Both versions have 229,317 trainable parameters; they differ in data transformation and sequence length.

Item	Settings used in this run
Inputs / features	42 date-ordered observations per stock: Open, High, Low, Close, Volume, ExpectedDividend. **42 is an observation count, not 42 calendar days.**
Convolution / recurrent layers	Conv1d: 6 to 64 channels, kernel 3, stride 1, padding 0; LSTM: hidden 100, 3 layers.
Dropout / output	Channel dropout 0.2; inter-layer LSTM dropout 0.2; dropout 0.2 at the last time step, then Linear 100 to 1.
Loss / optimizer	L1Loss (MAE); Adam lr 0.001, betas (0.9, 0.999), eps 1e-8, weight decay 0, amsgrad False.
Batch / epochs / seed	Batch 64; shuffle=False; fixed 20 epochs per fold, 160 total; seed=20261016.
State across folds	The same model and Adam optimizer continued training; these are not eight independently reinitialized experiments.
Actual entry point	Pinned model_2; the runner forced TIME_SPAN=[42]. model_1 in the uploaded main was not the entry point for this run; no early stopping.
Device / initialization	Baseline: CPU / torch 2.12.0+cpu; this run: CUDA cuda:0. Three smoke checks confirmed identical initial parameters and working CUDA forward/backward passes.

Version check: The current local JPX project has TIME_SPAN=21 and early stopping in its training function. Those settings cannot be applied retrospectively to this 42-observation, fixed-20-epoch run. This report uses the frozen version matching the run.

The platform listed T4 x2, but the log only establishes that the model ran on cuda:0; it does not establish dual-GPU use. The complete run_config.json was not retrieved, so CUDA, PyTorch, and other package versions are not filled in by assumption.

02 Data splits and evaluation definitions

The same source CSV, dates, and sample rules were used. Overall and fold statistics are treated separately. [B1-B3, S1-S3]

The dataset is stock_prices.csv (217,603,723 bytes). The local CSV SHA256, baseline metadata, and this run's log match exactly. Each expanding training interval starts on 2017-01-01. Validation intervals cover 180 calendar days, with an inclusive start and exclusive end. The final interval shorter than 180 days was not evaluated.

Fold	Actual evaluated Target dates	Days	Training samples	Evaluation samples
1	2017-06-30 to 2017-12-26	122	147,019	228,778
2	2017-12-27 to 2018-06-22	119	375,774	224,999
3	2018-06-25 to 2018-12-21	125	600,748	238,188
4	2018-12-25 to 2019-06-19	114	838,894	219,242
5	2019-06-20 to 2019-12-16	121	1,058,106	234,578
6	2019-12-17 to 2020-06-12	117	1,292,649	229,273
7	2020-06-15 to 2020-12-10	121	1,521,885	238,665
8	2020-12-11 to 2021-06-08	119	1,760,514	237,073
Total	2017-06-30 to 2021-06-08	958	Expanding intervals; not summed	1,850,796

Dates above come from the baseline's actual evaluation samples. This runner reused the same data and splits and checked each fold's sample count; the counts match the log exactly. Weekends and market holidays make actual dates differ from split boundaries; for example, Fold 4 first has samples on 2018-12-25.

Preprocessing and time boundaries

Missing ExpectedDividend values were first filled with 0. After valid rows were identified, prices were adjusted and filters applied. The scaler was fitted on each fold's training partition, then applied to validation. Standardization occurred before removing the final two training dates. Validation included up to 41 earlier observations per stock as input history. This run compared its mean/std with the baseline using a tolerance of 1e-8.

Metric calculations

Each day, stocks were ranked by prediction: the top 200 were long and the bottom 200 short. Raw weights linspace(2,1,200) were normalized so that each side's absolute weights sum to 0.5, for total gross exposure of 1. Daily r = Σ(weight x Target).

Sharpe = mean(r) / sample_std(r, ddof=1), with base=0, unannualized. The overall value is recomputed using all 958 days, rather than averaging fold Sharpes.

Rank IC = each day's cross-sectional Spearman(Prediction, Target), averaged over defined daily values; pandas mean excludes NaN.

Earnings = Σ daily r, before costs. **MAE** = mean absolute error across all samples. This run's overall MAE was reconstructed by weighting folds by sample count.

This is the project's offline historical evaluation, not an established Kaggle submission score. Prediction timing, ranking ties, data coverage, and submission format still require full submission validation.

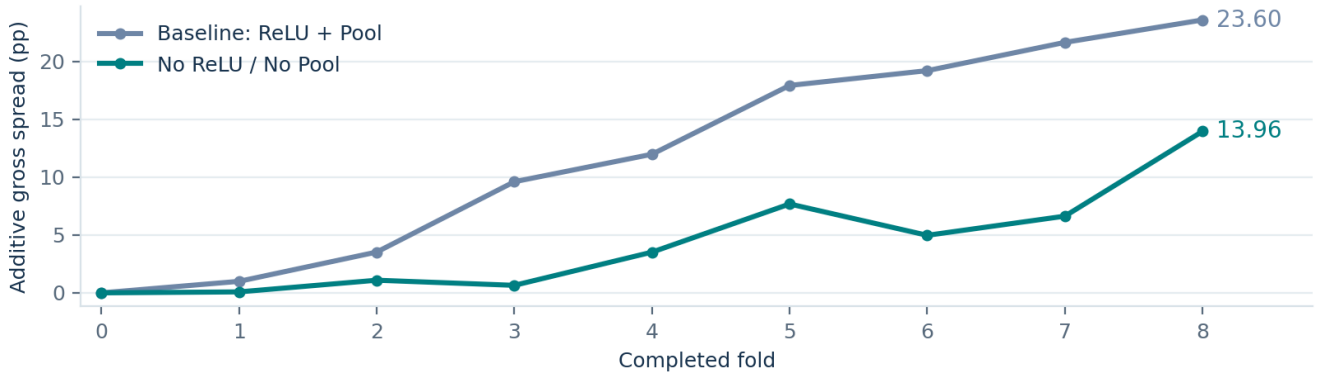
03 Results by fold

All values come from deduplicated FOLD COMPLETE records. MAE in this table is evaluation error. [L1]

Fold	Sharpe	Rank IC	Evaluation MAE	Earnings (pp)
1	0.004841	0.008864	0.012265	+0.0847
2	0.040872	0.000930	0.014818	+1.0078
3	-0.018476	0.004107	0.017379	-0.4412
4	0.120059	0.003253	0.015764	+2.8700
5	0.144857	0.008042	0.013384	+4.1628
6	-0.062112	-0.009502	0.022416	-2.7084
7	0.038872	0.003098	0.016873	+1.6571
8	0.188066	0.018991	0.015272	+7.3298

Overall pooled Sharpe = 0.054660544; mean daily Rank IC = 0.004774635. Six folds had positive gross spread and two negative; seven folds had positive Rank IC, although the values were generally close to zero.

Fold-end cumulative spread: later periods supplied most of the recovery



The chart connects cumulative values only at each fold end to compare period contributions. It is not a daily equity curve and cannot be used to calculate true daily maximum drawdown.

Three most informative findings

Fold 6 was weakest: Sharpe -0.062112, Rank IC -0.009502, MAE 0.022416, and gross spread -2.7084 pp. This period combined larger prediction error with ranking-portfolio losses. Dates alone cannot attribute the result to a particular market event.

Fold 8 had the best Sharpe, Rank IC, and Earnings: +7.3298 pp represented 52.50% of the run's total spread. The positive overall result is therefore sensitive to the contribution of the final period.

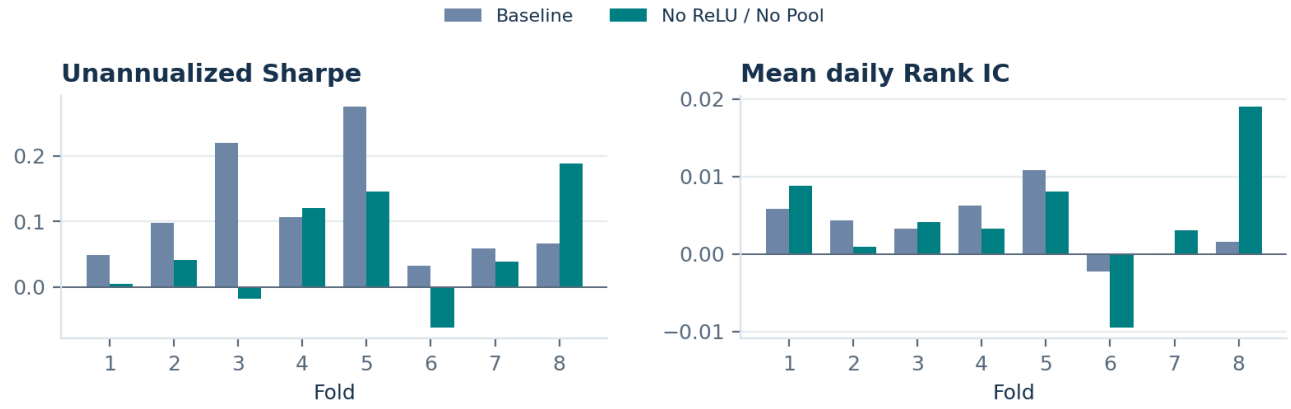
Fold 3 had positive Rank IC but negative Sharpe and Earnings. Rank IC measures ranking correlation across all stocks, while spread depends on the 200 stocks at each tail and their weights. These are different objectives.

04 Period-by-period comparison with the original

The original is the 42-observation CNN-LSTM with ReLU + MaxPool1d, not ENSEMBLE or the 21-observation CNN_ENSEMBLE. [B3, L1]

Fold	Δ Sharpe	Δ Rank IC	Δ MAE x 10^6	Δ Earnings (pp)
1	-0.044363	+0.003060	+7.797	-0.9124
2	-0.056522	-0.003445	+13.746	-1.5240
3	-0.238033	+0.000802	+13.990	-6.5215
4	+0.013396	-0.003041	-5.257	+0.4950
5	-0.129570	-0.002754	-31.410	-1.7867
6	-0.093897	-0.007262	+18.499	-3.9872
7	-0.019408	+0.003063	+10.236	-0.7855
8	+0.121859	+0.017434	-73.112	+5.3898

Δ = this run - original. Positive Sharpe, Rank IC, and Earnings changes mean higher values; only a negative MAE change means improvement. The MAE column is multiplied by 10^6 for readability.



Sharpe and Earnings exceeded the baseline only in Folds 4 and 8 (2/8); Rank IC was higher in Folds 1, 3, 7, and 8 (4/8); MAE was lower in Folds 4, 5, and 8 (3/8). This is not a broad, consistent improvement across periods.

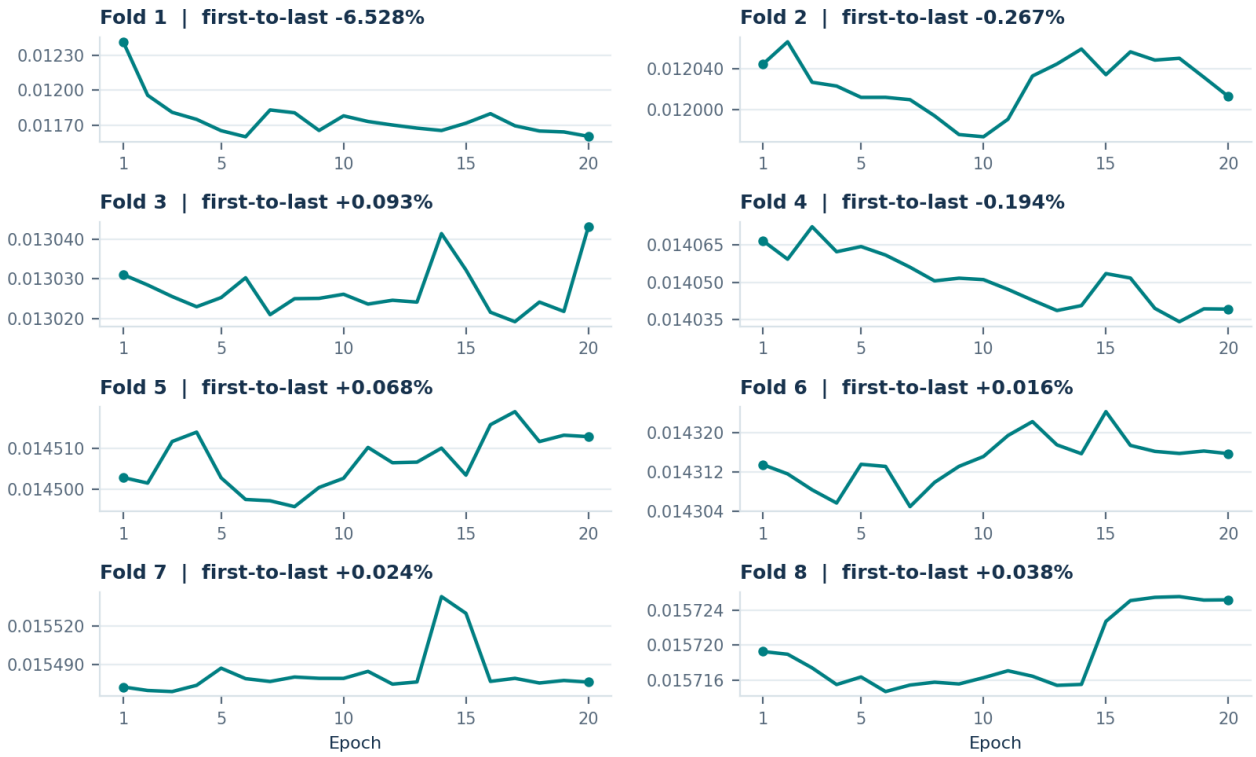
The largest gross-spread shortfalls came from Fold 3 (-6.5215 pp) and Fold 6 (-3.9872 pp). Fold 8 improved by +5.3898 pp, but this did not offset declines in other periods.

The overall Rank IC gain also depends on Fold 8. A sensitivity calculation using the first seven folds and the baseline's valid-day counts (assuming Fold 7 also has 120 days) gives 0.00406638 for the original versus a lower 0.00275584 for this run. This is not another independent test.

The baseline's weighted MAE was 0.016032527 versus 0.016026605 for this run, an absolute reduction of only 0.000005922. The same-data zero-prediction MAE, 0.015950026, remained below both models. Zero prediction is an error reference; its many ties prevent it from serving directly as a valid ranking-trading benchmark.

05 Training trends and execution efficiency

All 160 epoch records are included; duplicate logs do not inflate the training count. Each small chart has its own y-axis range. [L1, B3]



Fold	Epoch 1 MAE	Epoch 20 MAE	Minutes for 20 epochs
1	0.012415	0.011604	3.22
2	0.012045	0.012013	8.24
3	0.013031	0.013043	13.28
4	0.014067	0.014039	18.42
5	0.014503	0.014513	23.14
6	0.014313	0.014316	28.30
7	0.015473	0.015477	33.08
8	0.015719	0.015725	38.39

Fold 1's MAE fell by approximately 6.53% from the first to the last epoch; later folds changed little and did not always decline. Since the model and Adam continued across folds, epoch 1 in later folds inherited earlier training. Flat curves cannot directly be read as no learning or proof of convergence.

The log's training average loss is the average over all 20 epochs in a fold, not epoch 20 and not validation MAE. Each fold reports only final validation; there is no per-epoch validation curve, so the best validation epoch or the onset of overfitting cannot be identified.

Recorded epoch times total 2.768 hours in this run; the full main program took 2 hours 49 minutes 12.2 seconds. Completed baseline epoch times total 11.016 hours, an observed ratio of about 3.98x. CPU/CUDA, packages, and architecture differed, so this ratio does not measure acceleration caused by removing the components alone.

06 Anomalies, deduplication, and completion audit

Additional project evidence clarifies issues that the log alone could not resolve. [L1, B2-B3, S1-S3]

Fold 7 constant-input warning: a known data cause

> The baseline confirmed that all Targets on 2020-09-29 were 0: Spearman correlation was undefined for the 1,973 stocks that day. The original produced 1,973 distinct predictions, so its warning was not caused by identical model outputs.

Using the local CSV with the same hash, this report repeated the frozen row filters, Fold 7 split, history completion, and window-eligibility checks. It confirmed exactly 1,973 evaluated Targets that day, all 0. The runner first called the original evaluation function, then computed IC again to save daily results, so the same issue generated a warning in both calculations.

Weighting this run's fold ICs by the baseline's valid IC days (120 for Fold 7, the original day counts for other folds) exactly reproduces 0.004774635358040866, consistent with an average over 957 valid days. However, this run's daily prediction file has not been retrieved, so the absence of constant predictions or other additional degeneracy cannot be claimed.

Five snapshots merged into one complete run

Original line range	Records	Treatment
81-1279	1,199	Identical prefix of the final complete block; removed
1280-1603	324	Identical prefix; removed
1604-4216	2,613	Identical prefix; removed
4217-6924	2,708	Identical prefix; removed
6925-9688	2,764	Retained the complete run through HTML export completion

Timestamps and messages were compared, ignoring Kaggle display counters. A total of 6,844 records were removed from the first four blocks. The source has 9,688 lines, including 80 lines of page-interface text. Global line-by-line deduplication was not used, preserving smoke checks and warnings at different times and legitimate repeated blank lines.

Completion evidence and non-fatal warnings

At 10,184.6 seconds, Fold 8 completed and model saving was logged; at 10,184.8 seconds, Program finished and COMPLETED appeared; at 10,186.2 seconds, the results directory was listed; at 10,194.1 seconds, HTML export finished. All eight folds, 160 epochs, 1,850,796 samples, and every printed finite loss were checked.

The header still read Running / Output 0 B, contradicting the completion evidence at the end; it cannot establish that the run was unfinished. Frozen-module debugger warnings and mistune / nbconvert syntax warnings did not stop execution.

This report checked the completion log and runner output logic. The run's checkpoint, daily_metrics.csv, prediction parquet, and run_config.json have not been downloaded. Therefore, "the program reported saving files" and "the output files were locally verified as complete" remain separate statements.

07 Research conclusions and next steps

Supported conclusions, unresolved issues, and follow-up experiments are separated so that one comparison is not treated as a definitive architectural verdict.

Current best judgment: retain the baseline. Removing both components increased overall Rank IC but sacrificed more Sharpe and spread. Its MAE improvement is too small, with no consistent advantage across folds, to justify replacing the baseline.

Supported conclusions

Data hash, sample counts, and normalization are aligned; parameter counts and initialization are comparable. Under the same project evaluation definitions, this run increased Rank IC, reduced Sharpe and the gross additive spread sum, and lowered MAE only slightly. The hypothesis of "improving Rank IC and Sharpe simultaneously" was not supported by this observation.

Conclusions that remain unsupported

ReLU or Pooling cannot individually be judged harmful or beneficial: both were removed together, and the device changed from CPU to CUDA. A single seed, a model and optimizer carried across folds, and the same historical data path do not constitute eight independent repeated experiments. Without this run's daily series, credible paired-difference intervals, turnover, after-cost results, or daily drawdowns cannot be produced.

Despite the two-day purge and training-fitted scaler, not all temporal leakage risks can be claimed to be excluded. The scaler was fitted before purging; label timing for sparse stocks, stock boundaries, price-adjustment timing, and information availability at actual order time require a separate dedicated audit.

Recommended validation sequence

Step	Specific action	Question to answer
1	Retrieve this run's daily IC/spread, predictions, run_config, and final weights; check hashes and row counts.	Do warnings arise only from the known constant Target? Can the results be fully reconstructed?
2	Fix the device, packages, and data; compare four groups: original, ReLU removed only, Pool removed only, and both removed.	What are the individual and interaction effects of the two components?
3	Repeat all four groups with multiple prespecified seeds; save full daily series and training/validation curves.	Are the differences stable, or do they reflect one random and numerical path?
4	Align results by day; use a paired block bootstrap that preserves time dependence for exploratory intervals, and retain an untouched test period.	How uncertain are the historical differences, and do they persist across periods?
5	Add checks of ranking quality in the 200-stock tails, turnover, costs, and tradable timing.	Why did better Rank IC not translate into Sharpe, and is the strategy actually tradable?

These are follow-up research recommendations, not work performed in this report. In particular, the baseline was not retrained, and baseline turnover was not substituted for this run's turnover to estimate net performance.

08 Sources, definitions, and reproducibility scope

This report combines the complete training log, originally uploaded model, frozen code, and baseline result files. All charts were drawn from values that were obtained.

Source index

[L1] Pasted text.txt: 9,688 original lines. Original lines 6925-9688 were retained; the completion summary is at original line 9679, and the eight-fold list at line 9673. Deduplicated filename: jpx_no_relu_42day_deduplicated.txt.

[B1] Baseline experiment_config.json: data hash, six features, model/Adam parameters, date splits, scaler, and sample counts.

[B2] Baseline evaluation_protocol.md: metric formulas, gross-spread units, 957 valid IC dates, and diagnosis of constant Targets on 2020-09-29.

[B3] Baseline summary.json: raw metrics for eight folds, overall MAE/zero-prediction MAE, pooled metrics, and completed epoch times.

[S1] Full file-read output of run_no_relu.py and evaluation.py saved in the Kaggle submission task, from the corresponding "Kaggle Model Training" task record. Used to check the fixed window of 42, 20 epochs, CUDA, initialization/sample checks, outputs, and pooled metric formulas.

[S2] User-uploaded models(1).py and main(3).py: the former verifies the architecture without ReLU/Pooling; the latter is retained as the originally uploaded entry point. This run actually used the pinned model_2.

[S3] Frozen support code, codes/: model_2, features, data_processor, deep_learning, evaluation. The current local Desktop/JPX state was also read to identify later configuration changes.

Version and data identifiers

Item	Identifier
Baseline results revision	b128fd5c26e4a3a4d57cadb919c7582cd502eb5d
Support code revision	b3cd1bfd617251c7eebc42c588e4067a09af1fe3
CSV SHA256	bf774a86f834e5338bba74f2356ebb750f93a1cc4e89b064ee07251e8ed851db
Final model in this run	CNN_LSTM_20261020_133013_901214_c3e8d799.pt
Kaggle results directory	/kaggle/working/jpx_no_relu/results

Reconstructed calculations and precision

This run's weighted MAE = Σ(fold MAE x fold samples) / 1,850,796 = 0.016026604836824086; the sum of fold Earnings = 0.1396259912289679. The baseline total is 0.2359505807980895; summing its daily values in float64 gives 0.2359505802228341. The last-digit difference comes from the original fold-level float32 reduction.

Charts and difference tables use original precision and are rounded only for display. This run's Sharpe and Rank IC are taken directly from the completion summary. The original runner converted all daily spreads to float64 to calculate pooled Sharpe. MAE was reconstructed from the recorded batch-weighted averages; this run's individual predictions were not reread.

This report claims no guaranteed profits, statistical significance, official Kaggle submission score, or complete audit of daily output files. Data not obtained remain unverified; missing values are not filled by conjecture.